Program 2: Write a program using a function that returns the area and circumference of a circle whose radius is passed as an argument.two values using tuple assignment

```
pi = 3.14
def Circle(r):
    return (pi*r*r, 2*pi*r)
radius = float(input("Enter the Radius: "))
(area, circum) = Circle(radius)
print ("Area of the circle = ", area)
print ("Circumference of the circle = ", circum)
```

Output:

```
Enter the Radius: 5
Area of the circle =  78.5
Circumference of the circle =  31.400000000000002
```

Program 3: Write a program that has a list of positive and negative numbers. Create a new tuple that has only positive numbers from the list

```
Numbers = (5, -8, 6, 8, -4, 3, 1)
Positive = ( )
for i in Numbers:
    if i > 0:
        Positive += (i, )
print("Positive Numbers: ", Positive)
```

Output:

```
Positive Numbers:  (5, 6, 8, 3, 1)
```

9.3 Sets

Introduction

In python, a set is another type of collection data type. A Set is a mutable and an unordered collection of elements without duplicates. That means the elements within a set cannot be repeated. This feature used to include membership testing and eliminating duplicate elements.

12th Computer Science_EM Chapter 9.indd 156 21-12-2022 15:37:02